

Jackson Bean

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EDUCATION

The Ohio State University

B.S. in Computer Science and Engineering and Theoretical Mathematics

August 2021 - May 2026

Columbus, OH

GPA: 4.00/4.00

EXPERIENCE

Student Instructional Associate

The Ohio State University

August 2024 – Present

Columbus, OH

- Prepare and teach College Algebra recitation twice a week to over 70 students divided into two sections
- Answer over 20 questions each week about course material in a concise and understandable manner
- Prepare, grade, and proctor 10 weekly quizzes and four monthly exams over semester-long course content

Cognitive Systems Engineering Research Intern

The Ohio State University

May 2024 – September 2024

Columbus, OH

- Worked with the Cognitive Systems Engineering Lab (CSEL) to develop two new contingency plans for UAMs
- Developed C++ code to run more than 40 simulations that enacted parameter-determined flight pathing
- Plotted results of simulations on six dynamic charts and graphs using Python with Pyplot Dash

Biomedical Informatics Research Intern

The Ohio State University

May 2023 – August 2023

Columbus, OH

- Constructed and trained an AI model which classified comatose patient outcomes with more than 80% accuracy
- Programmed Python code to read and sanitize over one million instances of patient data
- Interpreted classification results using R and presented findings at a departmental poster session

PROJECTS

Molecular Dynamics Simulation (OSU Cycle) | *Python*

September 2025 – Present

- Develop algorithms to simulate a class of molecules and their related intramolecular forces
- Generate simulations to analyze molecular behavior under external force fields
- Approximate physical values using numerical methods for solving partial differential equations

Personal Portfolio Website | *React, TypeScript, Tailwind CSS*

June 2025

- Developed a dynamic personal portfolio website using industry-standard tools for front-end web development
- Implemented reusable stateful components to organize content for a cohesive user experience
- Hosted the website using GitHub Pages with a custom domain name

Full-Stack Employee Database Web Application | *React, TypeScript, Express.js, MongoDB*

March 2025

- Built a RESTful API backend to access a cloud-based database through MongoDB
- Applied CRUD functionality to a backend component through custom GET, POST, and DELETE routes
- Programmed and styled a front-end webpage to easily search the database via keywords

SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, Java, SQL, HTML/CSS, Ruby, x86 Assembly, R, LaTeX

Frameworks & Libraries: NumPy, pandas, scikit-learn, Express.js, Ruby on Rails, React, Tailwind CSS

Developer Tools: MongoDB, SQLite, Linux/UNIX, Makefile, Git, Vite, VS Code, Xcode, Postman, Docker

Mathematics Experience: Real Analysis, Abstract Algebra, Numerical Analysis, Graph Theory, Mathematical Statistics, ODEs, PDEs, Linear Algebra, Calculus

Miscellaneous: Personal Computer Building, Global Seal of Bilingualism in Spanish